Module: game engine

English version of ../../modules/game-engine.md.

1. Purpose and boundaries

Game rules: round generation, answer checking, scoring, modes (classic, learning, campaign). A domain layer without Flask and network. It does not send events to clients (that is `realtime`) and does not store the leaderboard (that is `services.leaderboard`).

2. Use scenarios

- Round start: a set of questions from the bank, a time limit.
- Player answer: checking, scoring, recording the result.
- Round end: totals are computed and handed to the leaderboard.
- Failure: not enough questions → a round configuration error, not an empty round.

3. Contract

Internal functions of `app/game_logic.py` and `app/domain/*`; the outside world sees Socket.IO events and `/api/*` responses. Error codes — `app/errors.py`.

4. Data

Reads the question bank (built-in/offline or AI-generated), writes the round result through the leaderboard service. Has no tables of its own.

5. Access rules

Game actions are available anonymously; the cheat mode (`/api/cheat`) requires `CHEAT_TESTER_CODE` and is checked separately (`tests/test_privileged.py`).

6. Dependencies

Outgoing: `app/domain`, `app/ai_client` (question generation, optional). Incoming: `app/http_api/rooms.py`, `app/realtime/socket_events.py`.

7. Configuration

`CHEAT_TESTER_CODE` (access to the test mode). Question count and round time come from the room data.

8. Build and debug

```
make test
python -m pytest tests/test_domain.py tests/test_services.py -q
```

9. Tests

`tests/test_domain.py` (rules and scoring), `tests/test_services.py`, `tests/test_validation.py` (input boundaries), `tests/test_privileged.py` (access to privileged actions).

10. Observability

Round and generation error logs; room state is visible through Socket.IO events. Errors: `invalid_answer`, `round_not_ready`, `privileged_required`.

11. Limitations and readiness

- Rules live in one process: several replicas will not share room state.
- Round state is not persisted: a restart interrupts the game.
- Readiness: the rules are covered by unit tests; the privileged mode is closed by a code.